

COLLISION Phase 14: Model Capacity Scaling

Quick Statistics

Dataset Version:	collision_dataset_v5_expanded (Train: 1,546,977 tokens)
Model Parameters:	10,282,304 (6 layers, 384 d_model, 8 heads, 768 d_ff)
Vocabulary Capacity:	8,000 capacity (weight-tied embeddings)
Steps Trained:	1,500 steps (from Random Initialization)
Hardware / Device:	CPU-compatible training run

Learning Metrics Progression (10M Model)

Step	Train Loss	Val Loss	Perplexity	Learning Rate	Speed (tok/s)
500	2.7877	2.6326	13.91	0.000500	803.5
1000	1.0413	0.9119	2.49	0.000593	876.6
1500 (Final)	0.3987	0.4824	1.62	0.000564	837.4

Model Capacity Comparison (3.38M vs 10M)

Metric	3.38M Base Model	10M Scaled Model
Test Loss (Lower is Better)	1.3213	0.7679
Test Perplexity (Lower is Better)	3.75	2.16
Average Repetition Rate (Lower is Better)	47.9%	26.8%
Average Unique Token Ratio (Higher is Better)	52.1%	73.2%
Sentence Termination Rate (Higher is Better)	55.6%	88.9%
Average Response Length	76.1 tokens	58.0 tokens

Generation Coherence Samples

Prompt: "Artificial intelligence is"

3.38M BASE:

to digned to sw crestion: to configain dual)

Answer: maller to s to locaus by and dectromagneticement...

10M BASE:

devel to maximize heatinforcement reward policies through reinforcement learning
policies where agents optimize behavior paths by maximizing rewards.

Prompt: "To prevent overfitting, a model should"

3.38M BASE:

igit: cas transform therve where of of curvital distributions is stand...

10M BASE:

ecision trees of to maximize information gain minimize Gini impurity values.

Learning Loss Curve

COLLISION-10M Pretraining Loss Curve

